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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Terrabotics
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDC 8 & SDC9
Project / Solution Name	SimpanAI
Name (Leader)	Lo Yu Ao
Email & Phone (Leader)	loaidan0608@gmail.com +60195310530
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

As traditional shopping shifts toward online platforms like Shopee and Lazada, Malaysian micro and small businesses (SMEs) struggle to manage inventory alongside daily operations. Unlike large corporations that can afford complex storage systems and dedicated warehouse teams, small businesses rely on staff using disconnected spreadsheets, paper records, and WhatsApp messages.

Without automated tracking or visibility into previous years' sales trends, staff frequently overlook depleting stock counts, leading to sudden stockouts and lost sales opportunities. Conversely, items left sitting in storage too long go unnoticed, causing food and perishable products to expire and hardware or electronics to degrade or malfunction.

This problem is serious for SME because inaccurate restocking drains SME cash flow and profit margins which impacts the business. Environmentally, expired perishables and damaged goods create avoidable material waste. Socially, owners and staff face burnout spending evenings on repetitive manual tracking instead of growing their business.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Built using Replit, our web application provides an intuitive inventory management and automated replenishment module for SMEs within the Orbyte ecosystem. Users can register custom products with specific safety stock limits and expiration dates, while viewing an overview of current storage counts and product statuses. Whenever users record completed sales, the system will automatically deduct the sold quantities from live inventory.

As stock levels change, the platform continuously monitors inventory health and automatically triggers alerts for low stock counts, aging stock, or approaching expiry dates. To eliminate manual guesswork, the system integrates AI predictive modeling to forecast future sales from historical transaction data and calculate optimal stock replenishment quantities.

To ensure people stay in control, the system requires explicit human approval before any AI-suggested replenishment is added to the inventory. Additionally, users retain full flexibility to manually purchase or input replenished stock whenever needed. By combining SDC Phase 1 skills demonstrated in the workshop, our solution prevents costly stockouts and expiration waste without requiring complex staff training.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

For our prototype, we used TypeScript, Vite, TailwindCSS, TanStack Query and Wouter to display the user interface where the inventory dashboard, expiry alerts and stock updates are shown. On the backend, we used TypeScript and Express 5 to process recorded sales, deduct stock counts, and check safety limits. SQLite was used as the database to store synthetic product records, expiry dates, and historical sales data. We also integrated RunPod to power the AI sales prediction model, while Replit hosts the entire website in the cloud.

By using these tools, we built an architecture that includes trigger, connected data, decision logic, human approval and visible outcome. First, a user logs a sale or enters a product on the TailwindCSS frontend, which sends a request through TanStack Query to the Express 5 backend. The Express 5 backend then updates the stock count inside the SQLite database and checks the item's safety stock limit and expiry date. When low or aging stock is detected, the backend sends past sales records from SQLite to the AI model running on RunPod, which predicts future sales and calculates a recommended replenishment amount. The suggestion is displayed on the frontend dashboard, and once the user clicks approve or manually purchases stock, Express 5 will update the final stock count and status in SQLite.

[Frontend: Vite + TailwindCSS + TanStack Query] ➔ [Backend: Express 5] ↔ [Database: SQLite] ↔ [AI Compute: RunPod GPU] ➔ [User Approval on Replit Dashboard]

Our website utilizes two official SDC sponsor platforms: Replit and RunPod. Replit was utilized in our project to build and host the website prototype in the cloud. Runpod was used to run the AI sales forecasting and provide replenishment suggestions to the user.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

To ensure our website is technically viable, the prototype can run reliably using Vite, Express 5, and SQLite on Replit without depending on live Orbyte access. The risk that a user might encounter when using this website is inaccurate AI prediction on RunPod when there is not enough historical sales data. To mitigate this risk, the website will refer to the safety-stock thresholds and manual stock entry to provide replenishment suggestions to the user so daily operations is uninterrupted.

Considering that many SMEs in Malaysia have varying levels of digital knowledge, our website is designed with a clean dashboard that requires zero formal training to understand. Instead of complex spreadsheets, staff see clear low-stock and expiry alerts, while human operators stay in control by approving AI replenishment suggestions or manually purchasing stock.

Our website is currently tested using synthetic data, and it is ready to accept real-world data in Phase 2. In a real deployment, users will need to set up their own account to log in the website which ensures the security of the website. A staff member can only use the website to check stocks and enter sales records, while the manager can decide on approval of AI suggestions and manual stock entry.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: Achieve higher levels of economic productivity through technological upgrading and innovation. Target 8.3: Promote development-oriented policies that support entrepreneurship and the growth of micro, small, and medium-sized enterprises (MSMEs). Targets 12.3: Halve retail/consumer food waste and substantially reduce material waste generation through prevention and automated expiry tracking
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Typical micro-SMEs relying on manual spreadsheets and WhatsApp spend an estimated 5 to 8 hours per week manually counting stock, checking expiry dates, and chasing supplier replenishment quotes. Small businesses also experience an estimated 10%–15% inventory loss from expired perishable goods or aging dead-stock electronics/hardware, alongside 15%–20% stockout rates on fast-moving items due to unpredicted demand.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction of expired or aging products discarded per month. Hours saved per week on manual stock-taking and replenishment calculations. % reduction in out-of-stock events and % accuracy of AI replenishment predictions compared to actual sales.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	40% reduction in expired perishable and aging stock waste through automated expiry alerts and AI demand-capped restocking. 70% reduction in weekly manual inventory management time. 50% reduction in unexpected out-of-stock incidents using RunPod AI sales forecasting and automated safety-stock triggers.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ■ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ■ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ■ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026